Foreign Direct Investment and Business Investment in Canada

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Abstract

Does foreign direct investment lead to more business investment in Canada? I examine this question using annual data from Statistics Canada and the World Bank for 1981-2018. I estimate a levels model with autoregressive errors and an error-correction model that includes expenditure, capacity slack, and FDI. In the main US-dollar models, the FDI coefficient is positive but statistically insignificant. The Canadian-dollar equations give a negative coefficient of about -0.107 to -0.108, including when the exchange-rate level is added. The result therefore depends on the currency and model used. The main US-dollar sample supports a cointegrating relationship between investment and expenditure, but serial correlation remains in the error-correction model. Overall, I find no robust evidence that FDI increases business investment. The estimates also do not show that FDI causes investment to fall. FDI inflows can finance acquisitions as well as new projects, so they should not be treated as a direct measure of additions to productive capacity.

1 Introduction

Foreign direct investment can help finance new plants, equipment, and technology. However, an inflow does not always represent new investment. A foreign investor may buy an existing company or replace financing that the company could have obtained elsewhere. Other firms may expand to supply foreign affiliates, or they may invest less as competition increases. These possibilities make the relationship between FDI and capital formation an empirical question.

I study Canada over 1981-2018, using business gross fixed capital formation as the dependent variable. The explanatory variables are gross national expenditure, estimated capacity slack, and FDI net inflows. Expenditure represents demand conditions, while capacity slack measures how far output is below estimated potential. Business investment includes spending by foreign-owned firms, so the outcome measures investment in Canada rather than investment by Canadian owners alone.

This paper revisits an earlier levels regression that found a positive FDI coefficient and compared it with one to assess complementarity. I reconstruct the data and reconsider that interpretation. Since investment and expenditure trend over time, I test for unit roots and cointegration before estimating their short-run relationship.

1 Introduction (continued)

The US-dollar models give positive but insignificant FDI coefficients. In Canadian dollars, the estimated association is negative and remains so after controlling for the exchange-rate level. I also extend the sample through 2024 and allow the FDI coefficient to vary with capacity slack. These checks show how sensitive the estimates are to the model; they do not establish a causal effect of FDI.

2 Background and investment model

Previous studies do not find the same investment response in every setting. Agosin and Mayer (2000) study developing economies over 1970-1996 and find crowding in in Asia and crowding out in Latin America. Borensztein, De Gregorio, and Lee (1995), using data for 69 developing economies, report a more-than-proportional relationship between FDI and total investment. The receiving economy and the way the relationship is estimated both matter.

Van Loo (1977) studies Canada and provides the main comparison for the earlier version of this paper. Van Loo reports an FDI coefficient of about 1.40, compared with less than 0.14 in my main US-dollar models. This difference cannot, on its own, show that complementarity has weakened over time. The samples, measures, and model specifications differ.

$$I_t = \alpha + \beta Y_t + \gamma U_t + \delta FDI_t + \varepsilon_t \quad (1)$$

In equation (1), I is business gross fixed capital formation, Y is gross national expenditure, and U is estimated capacity slack. I expect investment to rise with expenditure and fall when more capacity is unused. The expected signs are therefore positive for the expenditure coefficient and negative for the slack coefficient. The sign of the FDI coefficient is left to the data.

I also examine whether the association with FDI is stronger when spare capacity is available. The idea is that firms have more room to expand production when the economy is below potential; near full capacity, additional demand may instead raise prices. Because U measures slack, this hypothesis implies a positive coefficient on the FDI-by-slack interaction.

There are two reasons to interpret the coefficients carefully. First, expenditure includes investment, so the two variables are related through the national accounts. Second, FDI is a financial flow rather than a direct measure of new fixed capital. As Agosin and Mayer (2000) explain, acquisitions change ownership, and foreign firms can also finance investment locally. An FDI coefficient below one is therefore not enough to conclude that domestic firms are being displaced.

3 Data and methods

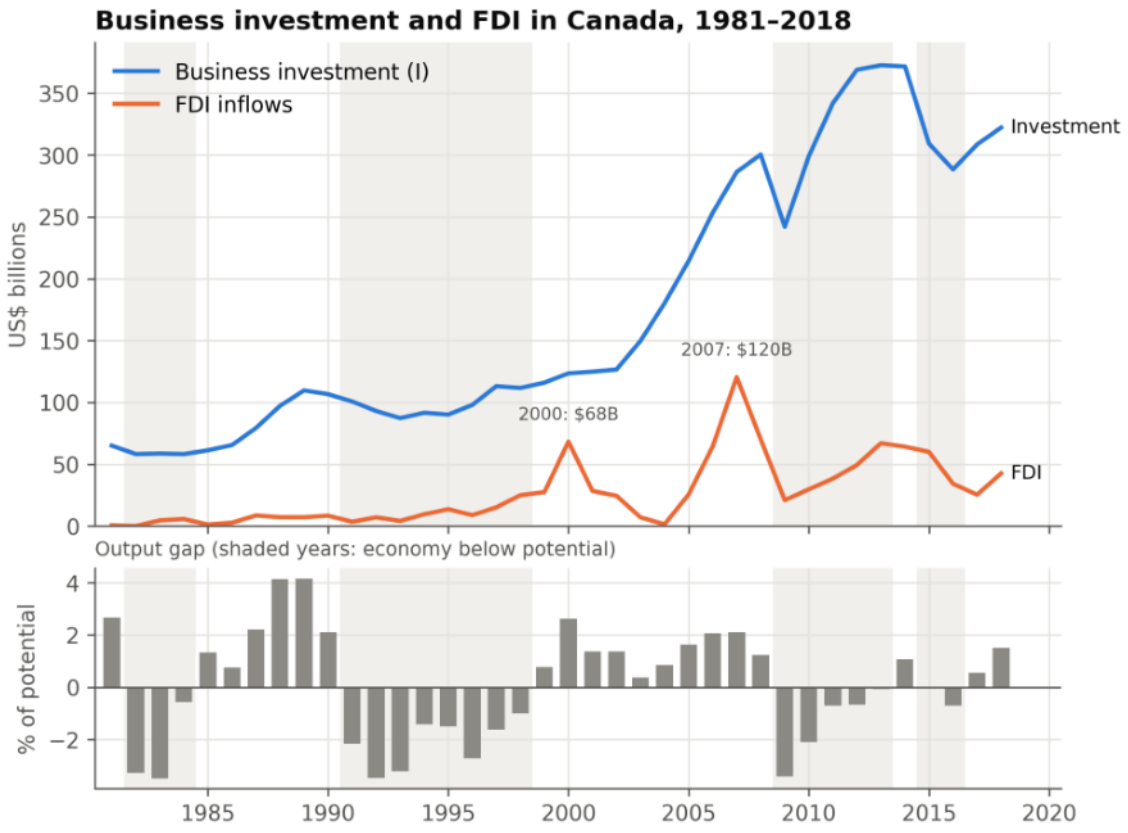
3.1 Data

The main sample has 38 annual observations, covering 1981-2018. I also use an extended sample ending in 2024. Statistics Canada Table 36-10-0222-01 provides business gross fixed capital formation, nominal GDP, and real GDP. The World Bank supplies gross national expenditure (NE.DAB.TOTL.CD), FDI net inflows (BX.KLT.DINV.CD.WD), and the annual average exchange rate (PA.NUS.FCRF).

3.1 Data (continued)

All monetary variables are in current-price billions. To obtain US-dollar values, I divide Canadian-dollar investment and capacity slack by the exchange rate, measured as CAD per USD. For the Canadian-dollar models, I multiply World Bank expenditure and FDI by that rate. There are no missing estimation values in the main sample. The earlier paper reports 36 observations, whereas the reconstructed sample has 38. Taking first differences leaves 37 observations; adding another lag leaves 36.

Figure 1. Business investment and FDI inflows



Sources: Statistics Canada Table 36-10-0222-01; World Bank WDI. Output gap from HP filter (λ = 100) on log real GDP.

3.2 Capacity slack

I estimate potential output by applying an HP filter to log real GDP over the full available series, using a smoothing parameter of 100. I then convert the log cycle into a proportional output gap and express the gap at the current GDP price scale. In equation (2), the trend term is the HP trend in log real GDP.

$$c_t = ln(GDP_{t,real}) - \tau_t \quad (2)$$

$$g_t = exp(c_t) - 1 \quad (3)$$

$$U_t = GDP_{t,nominal} \times [\frac{1}{1 + g_t} - 1] \quad (4)$$

A positive value of U means that output is below estimated potential. This differs from the earlier paper, which defined slack as potential output minus gross national expenditure.

3.2 Capacity slack (continued)

Although the HP filter removes an estimated trend from log real GDP, the final slack variable uses a changing nominal GDP scale. Its stationarity still needs to be tested. Using the full series also means that later observations and data revisions can change the estimated gap for earlier years.

3.3 Estimation

I estimate the models in Python using statsmodels. I begin with OLS in levels, then allow the errors to follow an AR(1) process using iterative GLSAR. This is different from the maximum-likelihood method used in the earlier EViews results. Allowing autoregressive errors addresses residual persistence, but it does not solve nonstationarity in the variables.

The augmented Dickey-Fuller tests include a constant and trend in levels and a constant in first differences. I select lag length by AIC. For the US-dollar series, the results support treating investment and expenditure as integrated of order one, and slack and FDI as stationary under these test specifications. I then use the Engle-Granger test for cointegration between investment and expenditure. The test rejects no cointegration in the main US-dollar sample ($p = 0.002$).

$$I_t = a + bY_t + v_t \tag{5}$$

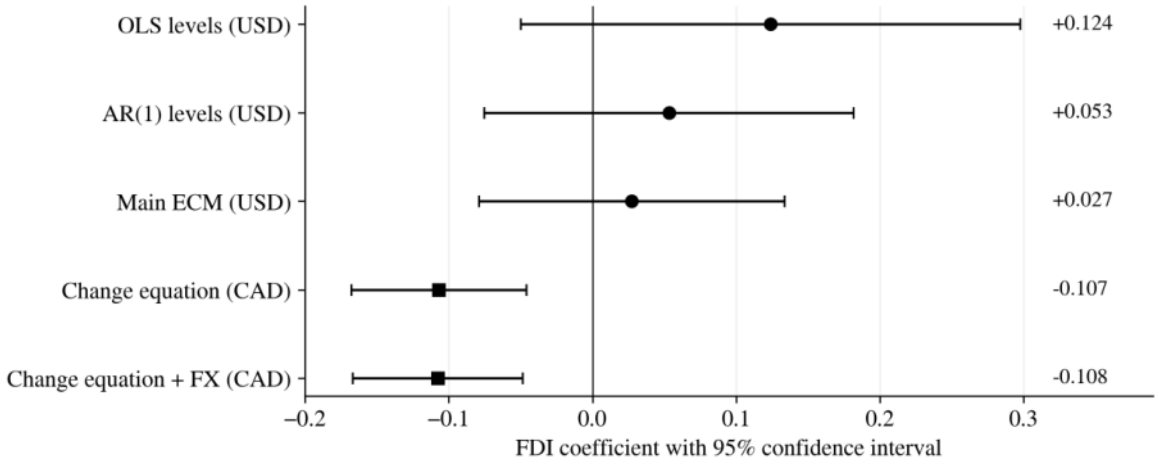
$$ECT_{t-1} = I_{t-1} - \hat{a} - \hat{b}Y_{t-1} \tag{6}$$

$$\Delta I_t = \kappa + \theta \Delta Y_t + \gamma U_t + \delta FDI_t + \lambda ECT_{t-1} + e_t \tag{7}$$

The error-correction term, ECT, measures how far investment was above or below its estimated relationship with expenditure in the previous year. A negative adjustment coefficient means that investment tends to move back toward that relationship. FDI and slack enter the short-run equation in levels; neither is included in the estimated cointegrating relationship. I use HAC standard errors with three lags. Other specifications add lagged changes, an FDI-by-slack interaction, or a 2020 indicator in the extended sample.

4 Results

Figure 2. FDI coefficients across specifications



Notes: Lines show 95% confidence intervals. Levels models use conventional standard errors; the main ECM and CAD change equations use HAC standard errors with three lags. FDI enters in levels throughout. Source: Author's calculations.

4 Results (continued)

At the 5 percent level, the ADF tests reject a unit root for slack and FDI, but not for investment or expenditure (Table 1). All four first-difference tests reject a unit root. These results support using an error-correction model, though the small sample makes the unit-root evidence less conclusive.

Table 1. Augmented Dickey-Fuller tests, 1981-2018

Variable	ADF stat (level)	p (level)	ADF stat (diff.)	p (diff.)
I	-2.99	0.136	-4.59	<0.001
Y	-3.33	0.061	-4.15	<0.001
U	-3.60	0.029	-5.41	<0.001
FDI	-4.99	<0.001	-5.25	<0.001

Levels include a constant and trend; differences include a constant. Lag length by AIC.

Table 2. Main US-dollar estimates

	OLS levels	AR(1) levels	ECM (HAC)
Y	0.194***	0.198***	
	(0.004)	(0.005)	
ΔY			0.224***
			(0.010)
U	-0.281***	-0.342***	-0.174*
	(0.096)	(0.078)	(0.099)
FDI	0.124	0.053	0.027
	(0.085)	(0.063)	(0.054)
ECT			-0.369**
			(0.158)
Intercept	-11.980***	-16.757***	-2.592*
	(2.952)	(5.784)	(1.345)
R-squared	0.994	0.983	0.923
R-squared Adj.	0.993	0.981	0.914
N	38	37	37

Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. ECM uses HAC (3 lags).

The FDI coefficient is 0.124 in OLS and falls to 0.053 after allowing AR(1) errors. Neither is significant at 5 percent. In the main ECM, the coefficient is 0.027, with a HAC p-value of 0.615. Across these US-dollar models, I cannot reject a zero FDI coefficient after controlling for the other regressors.

4 Results (continued)

The ECM adjustment coefficient is -0.369 (p = 0.019). Holding the short-run regressors fixed, this suggests that about 37 percent of the previous year's deviation from the estimated investment-expenditure relationship is corrected within a year. Investment changes are positively associated with expenditure changes, with a coefficient of 0.224. The slack coefficient is -0.174 (HAC p = 0.078), which has the expected sign but is not significant at 5 percent.

5 Robustness and limitations

Table 3. Sensitivity of the FDI estimate

Specification	N	Coint. p	FDI coef.	FDI p	FDI x U
ECM, US$ (main)	37	0.002	0.027	0.615	-
ECM + FDI x U	37	0.002	0.071	0.356	0.003 (p=0.267)
ECM, C$	37	0.382	-0.071	0.009	-
ECM, US$ to 2024 + COVID	43	0.358	0.032	0.548	-
First differences, C$	37	0.382	-0.107	<0.001	-
First diff., C$ + exch. rate	37	0.382	-0.108	<0.001	-

All models use HAC standard errors (3 lags). Coint. p = Engle-Granger test of I and Y.

The negative Canadian-dollar result needs separate consideration. Removing ECT from the CAD equation gives an FDI coefficient of -0.107 ($p < 0.001$). Including the exchange-rate level leaves it almost unchanged at -0.108 ($p < 0.001$). In the latter equation, annual inflows that are C$1 billion higher are associated with a C$0.108 billion smaller annual change in business investment, holding the other regressors fixed. FDI enters in levels, so this is not the estimated response to a C$1 billion year-to-year increase in FDI.

Adding the exchange-rate level does not reconcile the CAD and USD results. Converting currencies using a rate that changes each year changes the dynamics of the series. An additive exchange-rate control does not reproduce that transformation. The CAD variables retained in levels also need their own stationarity tests. The currency comparison is therefore a check using a different specification, rather than a simple change of units.

There is also unresolved serial correlation in the main ECM. The two-lag Breusch-Godfrey test rejects no serial correlation (p = 0.001) in the 37-observation sample. HAC standard errors do not fix omitted dynamics or endogeneity. In addition, business investment includes foreign-owned firms, expenditure includes investment, and the estimated output gap depends on the data vintage. Together, these issues limit what the estimates can tell us about displacement of investment by domestic firms.

The next steps are to test the CAD series separately, examine a small number of additional lags, and compare alternative measures of capacity slack. These checks would help assess whether the negative CAD association survives a better-specified dynamic model.

6 Conclusion

I find no statistically significant association between FDI and business investment in the main US-dollar models for Canada over 1981-2018. The Canadian-dollar change equations produce negative coefficients of about -0.107 and -0.108, including after adding the exchange-rate level. That negative association is worth investigating, but it is not consistent across currency specifications. Serial correlation in the main ECM also limits confidence in its estimated short-run dynamics.

The results do not provide robust evidence that FDI crowds in business investment, and they do not establish that FDI causes investment to decline. Aggregate inflows combine transactions with different economic effects. Financing a new production facility adds capacity, while buying an existing business may mainly change its ownership. The data used here do not separate those activities.

For policy, the amount of FDI received is therefore only part of the question. We also need to know what it finances and how other firms respond. Investment data separated by ownership, together with FDI data that distinguish new projects from acquisitions, would help explain whether the negative CAD association reflects displacement, acquisition activity, or economic conditions affecting both variables.

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